

REPTILES

PHYLUM	Chordata
CLASS	Reptilia
ORDERS	4
FAMILIES	89
SPECIES	About 10,450

Reptiles are egg-laying vertebrates that have a tough skin with a covering of scales. These ectothermic (cold-blooded) animals cannot generate internal heat. There are 4 orders of reptiles: snakes, amphisbaenians, and

lizards (collectively known as squamates); crocodiles, alligators, and caimans (crocodilians); tortoises and turtles; and tuataras. Most reptiles, including those that live mainly in water, lay eggs on land. The young emerge fully formed without a larval stage.



SCALES
The Gila monster belongs to the largest group of reptiles, the lizards. Unlike amphibians, from which they are descended, all reptiles have scales rather than smooth, moist skin.

Evolution

Reptiles first appeared about 340 million years ago, having evolved from early amphibians. The first reptiles differed from their amphibian ancestors in 2 significant respects: they developed a hard, scaly outer skin that protected them from abrasion and moisture loss; and, more importantly, they evolved a shelled, amniotic egg, in which the embryo developed within a sac of water, protecting it from the environment. Together, these features allowed reptiles to move away from the margins of water bodies, to which the amphibians were restricted, and to colonize land. By the Mesozoic era

(230–65 million years ago), reptiles had diversified into an enormous variety of types, and were the dominant land animals. The turtles and tortoises branched off at a relatively early stage, and by 200 million years ago, species similar to those we would recognize today were already present. Some time after this, reptiles began to diversify explosively. Among the orders that appeared at this time were the pterosaurs (flying reptiles) and the dinosaurs, along with 2 orders that survive today, the crocodilians and the tuataras. Later, another evolutionary line led to the appearance of the squamates—the lizards, amphisbaenians, and

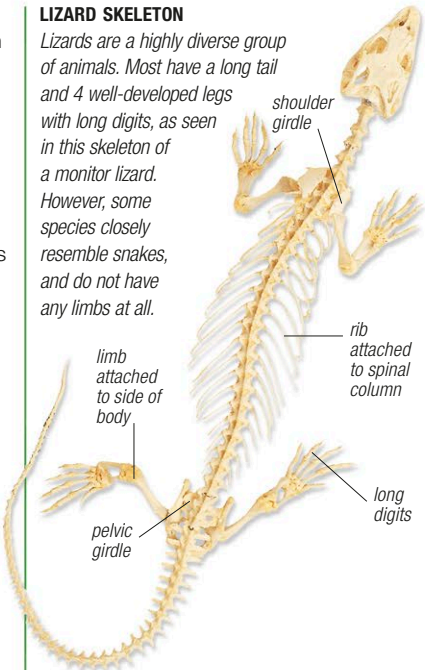
eventually the snakes. Of the 20 or so orders of reptiles known to have existed in the Mesozoic Era, only 4 survive today.

Anatomy

The external anatomy of reptiles shows immense variation, from the long, slender, limbless forms of snakes to the short, stout, shell-covered bodies of turtles. However, all reptiles are characterized by the presence of scales (see panel, below), which form a barrier that protects them from abrasion, attacks from predators and parasites, and dehydration. Scales differ considerably between reptiles, and in some species, scales of various shapes and sizes are found on different areas of the body. Pigment below the scales gives the animal its coloring and markings, which may be for camouflage or for display. In some reptiles, especially male lizards, groups of scales have evolved into dramatic crests, horns, and other features used for display.

LIZARD SKELETON

Lizards are a highly diverse group of animals. Most have a long tail and 4 well-developed legs with long digits, as seen in this skeleton of a monitor lizard. However, some species closely resemble snakes, and do not have any limbs at all.



The internal skeleton of a reptile is made up of bony elements and provides a stiff supporting system that is considerably sturdier than that of an amphibian, making reptiles more suited to life on land. Unlike those of mammals and birds, a reptile's limbs support the body from the side, which gives them a sprawling gait when they move. Snakes, along with most amphisbaenians and some lizards, are without functional legs.

Differences in the skull reflect the evolutionary origins of the various reptile groups and are a useful feature for classification. Turtles and tortoises have no openings in their temples, while crocodilians and tuataras have 2 on each side. The squamates also have 2 per side, but they are sometimes joined to form a single, large opening. In many reptiles, bone growth does not



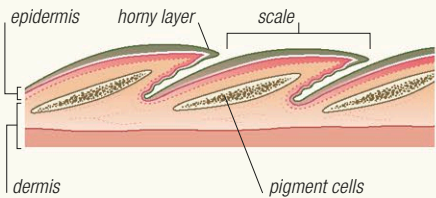
TURTLE FOSSIL
These are the fossilized remains of a turtle, dating back about 200 million years. Turtles and tortoises have changed little since that time, and are the oldest group of living reptiles.

SKIN STRUCTURE

A reptile's skin consists of 2 main layers: the epidermis (outer layer) and the dermis (lower layer). The scales, which are present only on the epidermis, are made of a horny substance, keratin, which is similar in composition to human hair and fingernails. The dermis contains nerves, blood vessels, and cells that support and nourish the epidermis. Unlike a fish's scales, those of a reptile cannot be removed individually. All reptiles replace their scales by shedding their outer skin. This allows room for growth, and also replaces skin that is worn out. Whereas snakes slough their skin in one piece, lizards, crocodilians, turtles and tortoises, and tuataras shed it in chunks or flakes. After a snake has shed its outer skin, it often appears much more brightly colored.



SCALE TYPES
Reptiles' scales differ greatly in size, shape, and texture. They may be smooth or rough, and may overlap, like roof tiles, butt up against each other, or have layers of stretchy skin in between. In crocodilians, the scales on the back are strengthened by bony plates.



CROSS SECTION OF SNAKE SKIN
The scales of a reptile are made from the thick, horny outer layer of skin. Each scale is joined to its neighbors by flexible, hinged areas, so that the body can move and bend. Pigment cells between the epidermis and the dermis determine the animal's coloration.